

brains in this niche field and bring them together under our banner to become the world's biggest cloud security company. Having said that, I am hopeful that the future of cloud security in India is pretty bright.

Q India is going online and there is growing awareness about security. But are Indians, particularly the SMEs and the SMBs, aware of the concept of website security?

India is going online at a faster pace than we expected and there is a lot of scope in that domain as well. Hundreds of e-commerce start-ups are being launched, which is a clear indicator of the growth in the online activity of Indians. Overall, it is a good scenario, but one cannot deny the fact that growing online activity also increases the threats. We at *Defencely.com* are serious about this issue, and we aim to be involved particularly with SMBs and SMEs to create awareness about the virtual losses that they may have to face because of not securing their websites. Let's imagine a scenario where an e-commerce portal gets compromised because of some zero day vulnerability, and hackers misuse the entire database of the portal, including the financial information and personal details of its buyers. Despite being technically sound, small and big online companies overlook the importance of online security. To a lot of online companies, the term 'hacking' means email or credit card hacking. But hacking has evolved and is a million dollar business at present. It is high time that Indian SMBs and SMEs get aware of such threats and start taking appropriate Web security measures. Modern day security services are complete packages of security.

Q What are the most common vulnerabilities that websites are exposed to, in the modern day world?

Some of the most common threats to most Web applications are:

- (a) Injection vulnerabilities and cross-site scripting: These are two of the most commonly found vulnerabilities that can occur in any website or Web application. There are various forms of injection attacks, including SQL, operating system, email and LDAP injection, and they all work by sending malicious data to an application as part of a command or query.
- (b) Cross-site scripting (XSS): These attacks target an application's users by injecting malicious code—usually client-side scripting such as JavaScript—into a Web application's output. Whenever the compromised output or page is viewed, the browser executes the code, allowing an attacker to hijack user sessions, redirect the user to a malicious site or simply deface the page. XSS attacks are possible within the contents of a dynamically generated page whenever an application incorporates user-supplied data without properly validating or escaping it.
- (c) Broken authentication and session management: Application functions related to authentication and session management are often not implemented correctly,

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allowing attackers to compromise passwords, keys and session tokens, or exploit other implementation flaws to assume other users' identities.

- (d) Insecure direct object references: A direct object reference occurs when a developer exposes a reference to an internal implementation object, such as a file, directory, or database key. Without an access control check or other protection, attackers can manipulate these references to access unauthorised data.
- (e) Security misconfiguration: Good security requires having a secure configuration defined and deployed for the application, frameworks, application server, Web server, database server and platform. All these settings should be defined, implemented and maintained, as many are not shipped with secure defaults. This includes keeping all software up to date, including all code libraries used by the application.

In order to prevent all these threats, developers need to have specific knowledge of how to code Web applications, keeping Web vulnerabilities in mind.

Q Can you share some tips to ensure website security?

It is sad to see websites getting hacked, by and large, on a daily basis. There are numerous steps to ensure a website's security. When was the last time you moved your files from their default locations to new ones? When was the last time you moulded your HT Access file, or when was it that you noticed your Web mail ID generating spam mail? When was the last time you scanned your website to check if it was affected with any new online threats? Such factors, among many others, play a vital role in shaping your website's security. I guess the best way to get started is to hire an online security company. There are some companies that don't charge a lot to trial-level users.

Q How is Defencely contributing to increasing awareness about website security?

Defencely doesn't limit itself to just providing cloud security services, but has a lot more to offer. Timely reporting of security vulnerabilities is what the core team at Defencely aims at. PayPal's recognition of Defencely being one of the top 10 cloud security companies isn't the only achievement of the firm. Several other giants such as Facebook, Apple, Zynga, SoundCloud and iFixit have applauded Defencely for its dedicated work.